

CHS — Handover Notes

Compiled: August 25, 2026, last updated October 9, 2026 — covering the full engagement to date, kept current alongside every batch of work, the same discipline already applied to the Users Guide, Terms & Conditions, and Feature Catalog.

Companion document: CHS_PROGRESS_LOG.md — a chronological record of what was built, in order. Read this file first for orientation, then the log for the detailed sequence.

1. What CHS Is

Complete Housing Solutions (CHS) — a Nigerian real estate PWA covering the full property journey: buying, renting, leasing, shortlet booking, and property sales, currently focused on Kaduna State. CHS is a genuine facilitator, not a passive listings board — it verifies documents and identities, holds funds in real escrow, and runs its own dispute-resolution process.

2. Stack & Repository

- **Frontend:** Next.js 16 (App Router), TypeScript, Tailwind CSS — repo folder `chs-v2/`
- **Backend:** Supabase (Postgres + Auth + Storage + Edge Functions), project ref `havwhdgjqgtxtqklkqfm`
- **Payments:** Paystack — currently on a **test key** (`pk_test_...`), not yet switched to live
- **Hosting:** Netlify, connected to GitHub for auto-deploy (confirm this is still true before assuming a push auto-deploys — it was manual drag-and-drop earlier in the project's history)
- **Migrations:** `backend/` (01–10, original schema) and `backend-v2/` (11–105, everything since) — sequential, numbered, meant to be read and applied in order
- **Documentation:** `CHS_USERS_GUIDE.docx` (+ in-app `/guide`), `CHS_COMPLETE_FEATURE_CATALOG.pdf` (+ in-app `/admin/feature-catalog`, + segmented `CHS_FEATURE_CATALOG_SEGMENTED.xlsx`), this handover note, and the progress log

3. Real Architectural Patterns Established

A new agent should follow these, not reinvent them:

- **RLS via `staff_can_access(domain)`** — every admin-facing table checks a real staff domain (`finance`, `customer_care`, `registration_setup`, `owner_buyer_tenant`, `agent_relations`, `artisan_dev_pm_vendor`), not a blanket "is admin" check. `finance` is deliberately not assignable to any sub-admin — Super Admin only, by design.
- **High-stakes actions go through a real approval queue**, not a direct write. See `request_admin_action()` / `apply_admin_action()` / `resolve_admin_action()` in `backend-v2/51_subadmin_roles_and_approval_queue.sql` onward. Anything financial or verification-related must route through this, not a raw table update.
- **Test data is always explicitly marked**, never silently mixed with real data. Pattern: `is_test_grant` boolean columns, `TEST-` prefixed references, and description text starting with  `TEST FUNDING`. Follow this pattern for any future test data.
- **Real, not simulated, verification.** Bank accounts are checked against Paystack's real account resolution API before they can be linked (`resolve-bank-account` Edge Function) — never trust a self-typed account name again.
- **Money-moving Edge Functions never trust client-supplied amounts.** Real fees are read server-side from `platform_settings`, not passed from the browser.
- **Apply every migration directly to the live database as it's built**, and test with real data (using `set_role_authenticated + set_config('request.jwt.claims', ...)` to simulate a real user's RLS context)

before considering it done. Supabase is always current; GitHub/Netlify only reflect the latest zip once actually pushed — these are two separate things on two separate schedules.

- **Always save every migration as a real local file in backend-v2/, the same moment it's applied to the live database — never apply-only.** A real, serious gap was found and fixed: migrations 77 through 95 were applied and tested live but never saved as files, meaning the exportable project and the live database quietly drifted apart for several rounds of work. Verify the file count directly inside a packaged zip before calling a package complete — don't assume the fix worked.
- **An "approve/confirm/mark paid" action must trigger its real consequence, not just update a status.** A comprehensive audit found 7 separate real instances of the same bug: a status changed and a notification fired, but no money, role, or state actually moved. Whenever building or reviewing any approval-style action, explicitly ask "what real thing is supposed to happen now, and does this code actually do it" — then prove it with a real before/after balance or state check, not just a read of the code.
- **When copying a "finished" documentation file between folders, always copy FROM the just-edited version TO the other location — never the reverse.** A real mistake happened doing exactly this backwards: freshly-updated Progress Log and Handover Notes were overwritten by their own stale originals from a different folder. Verify file content after any cross-folder copy, not just that the copy command succeeded.

4. Known Gaps — Read Before Assuming Something Works

Fixed since the last version of this document (kept here briefly so a new agent doesn't waste time re-discovering these): identity verification self-certified without real admin review; mobile logout was only reachable from the homepage; House Rules appeared on Sale listings where it made no sense; the "similar properties" advertising system had a real geographic-matching bug; sellers had no way to attach a real message when responding to an offer; the rent countdown and 90-day non-renewal notice had never been built despite the real database column existing; **most significantly, there was no real mechanism anywhere for a buyer to actually pay a seller the purchase price of a property** — only commission had ever been tested. All of these are detailed in CHS_PROGRESS_LOG.md.

Also fixed since the last version, Round 8: identity verification rebuilt to actually collect and check a name, state, gender, age bracket, address, occupation, email and phone, with a database-level lock so it cannot be forgotten on any single screen; rent now follows the same real escrow protection sale and shortlet money already had; the mobile-viewport bug — genuinely confirmed resolved this round after real-device testing, closing an item that had been open since the previous handover; the splash-screen/reload complaint, open for many rounds, traced to its real, final cause (a notification link opening a literal new browser tab) and fixed decisively.

Still genuinely open:

- **backend-v2/ is significantly out of sync with the live database.** Migrations 196–382 contain only summary comments, not runnable SQL — attempting to rebuild the database from this folder alone would fail for that range. This session's own files (383–410) were corrected; the older backlog was not, for lack of a safe, efficient way to pull ~900KB of historical SQL through a single conversation without crowding out the actual work being asked for. The live Supabase project remains the authoritative record in the meantime. This needs a dedicated pass — likely a proper pg_dump/migration-history export, not further hand-transcription.
- **Owner data-protection fix is only half-applied.** Owners could previously read an applicant's or buyer's full raw record through the API — NIN, ID document link, both phone numbers, the guarantor's ID and confirmation token — even though the on-screen display had been trimmed down earlier. Protected views now exist and the owner screens use them, but **the old, unrestricted direct database access is deliberately still open**, pending confirmation from the client that the owner screens work correctly on this release. Once confirmed, two policies need dropping (see migration 401_403's own header comment for the exact commands).
- **A real secret is embedded directly in the send-multichannel-reminder edge function's source** (INTERNAL_SECRET), and was exposed in a tool result during this round's audit. Treat it as compromised —

rotate it in the Supabase dashboard and update both the function and its one caller (`dispatch_due_reminders`).

- **A Privacy Policy now exists** (`components/PrivacyContent.tsx`, page `/privacy`; updated October 9, 2026 for wallet security data and mortgage handover details). It has NOT been reviewed by a Nigerian lawyer, and neither have Terms Version 6. Get both reviewed before treating them as final.
- **The daily automatic rent-release job (`release_due_rent`, migration 410) is newly built and has not yet been observed completing a full real cycle in production.** Tested directly against constructed data (a clean tenancy released correctly; an open fault correctly kept rent held), but a real, live 24-hour cron run has not yet been confirmed.
- **The company website (`website/index.html`) is not live.** No domain, no working email, several dead links, and its "Open the app" button points at an old Netlify address. It also asserts things (escrow, verification) that are now more true than when it was written, but should be re-checked against the current feature set before publishing. Two different company names appear in different places (Stadt - Thélina in the website footer, Complete Comprehensive Concepts, BN 3568074 on the in-app About page) — needs a single, deliberate decision on which is correct where.
- **No real recurring billing exists for any promotion package** (Classic/Premium/Elite/Signature), including Signature specifically. The database rows and pricing exist; the actual Paystack subscription charge was never built. Anyone assigned one of these packages today is effectively getting it free.
- **Pre-launch admin test mode** (Admin → Overview → "Switch role for testing") and the **Construction Roadmap test-unlock button** are both explicitly temporary. Remove both — and the underlying `grant_roadmap_access_test()` function — before real launch.
- **A real, substantial set of demo accounts exist in the live database**, built specifically so the client could manually test negotiation → payment → commission across categories, plus a second, newer Round 2 batch (0812... phone range). These are clearly named but are real, fully-verified, functioning accounts — clean up or keep, but know they exist before launch.
- **True server-side pagination was never built.** The homepage and admin queues are capped (e.g. `.limit(300)`), not genuinely paginated — fine for now, will need real work as the catalog grows.
- **Paystack is still on a test key.** Real payments do not move real money yet.
- **No custom domain configured.** Still on the default Netlify subdomain — note that WebAuthn's `WEBAUTHN_RP_ID`/`WEBAUTHN_ORIGIN` secrets will need updating to match whenever a real domain is set up, since WebAuthn ties strictly to the exact origin.
- **WebAuthn's domain configuration could not be directly verified.** The cryptographic implementation itself is genuinely solid, but secret values can never be read back once set — only overwritten. If biometric login ever seems to fail for every real user, this is the first place to check.
- **Real architectural design generation does not exist**, and was explicitly confirmed not to be a realistic near-term build — it would require either specialized generative-design/CAD software or real, licensed architects, not standard web development. Engage CHS's document delivery is a real human-upload tracker, not a generator.
- **Database backups / point-in-time recovery on Supabase — never confirmed.**
- **No uptime or error monitoring** — an outage today would be discovered from a user complaint, not an alert.
- **13 of the original 16 app views were never systematically compared against the original design spec** (a note from the very first handover, still open).

5. Major Systems Built Since the Last Version of This Document

- **Reminder Engine** (backend-v2/69) — one real, generic escalation system: rent-due cadence, maintenance follow-up escalation, and shortlet guest-message escalation, all sharing the same infrastructure. Multi-channel (in-app always works; SMS via Termii and email via Resend depend on real provider secrets being configured).

- **Estate Management** (backend-v2/70, 77, 78) — Estates as a real entity distinct from individual property delegation, with real service charges, bulk CSV unit onboarding, and a real subscription model (₦20k/50k/110k monthly by unit-count tier, confirmed with the client).
- **The real, complete commission model** (backend-v2/75, 76, 91) — Sale (6.5%/6%), Rental (5%/5.5%), Shortlet (real sliding scale by length of stay), Hire/Booking flat tier (Hotel/Event Centre/Car park), and Rent to Own/Mortgage (5%/5.5% per installment) — see the actual Terms & Conditions text (components/TermsContent.tsx) for the definitive, current numbers, since this is the one place they must never drift out of sync.
- **Rent to Own / Mortgage** (backend-v2/91, 94) — a genuinely new listing category, not the same thing as "Hire." Real installment payments, real ownership-percentage tracking, automatic conversion to a completed sale at 100%.
- **Shortlet/Hotel guest-host messaging** (backend-v2/92) — real in-app communication with genuine host anonymity toward the guest, never toward CHS itself.
- **Maintenance Reserve, now genuinely functional** (backend-v2/95) — real external funding, real artisan-payment draw order (reserve first, main wallet for any shortfall), real unrestricted withdrawal.
- **The real Sale purchase-price payment mechanism** (backend-v2/101, 102) — the single most significant fix of the engagement. Previously, only commission had ever been built or tested for a Sale; there was no way for a buyer to actually pay a seller. Now a single, transparent checkout: buyer's total automatically includes their own commission (shown as both percentage and real value), seller's net automatically excludes theirs.
- **Real, sourced legal document requirements and an escrow hold for Sale proceeds** (backend-v2/103, 104, 105) — Certificate of Occupancy, Deed of Assignment, Survey Plan, Governor's Consent, Tax Clearance Certificate, Sale Agreement, and conditionally Building Plan Approval, sourced from genuine Nigerian legal references. A buyer's payment is blocked until every required document is CHS-verified; a seller's proceeds are held (visible, not withdrawable) until CHS confirms the real, physical legal transfer is complete.
- **Real, admin-reviewed buyer identity verification** (backend-v2/96–98) — previously self-certified the instant the form was filled in; now a genuine pending → admin-review → approved flow.
- **Rent countdown and 90-day non-renewal notice** (backend-v2/99) — a real, live day-count on the tenant dashboard and a genuine notice mechanism, built after confirming the underlying database column had existed unused since early in the project.

New Since Last Handover Update (migrations 107–191)

- **Agent/Manager business tools** (145–149, 154–158, 175) — a real team/staff system (invite, role labels, daily reports), tenant and property/owner registers, expense tracking, owner remittance, auto-generated digital receipts, an independent per-owner commission model, and multiple bank accounts.
- **Host and Guest as real, separate roles** (160, 178–182) — split from Owner/Buyer with their own dashboards. The entire shortlet/hire payment mechanism was rebuilt: real request-to-book (guest sees full cost including commission before committing), a genuine host accept/decline, automatic refund on decline, and admin-released host payout — previously, hosts had no way to ever get paid at all.
- **Ratings, cancellation, and security deposits** (183–186) — real two-way ratings after a confirmed stay; a stated, server-enforced cancellation policy (100%/50%/0% by time-to-check-in); a flexible, host-toggled security deposit that auto-waives after a guest's 3rd real rating, resolved by admin (release or claim) rather than left to either party's word.
- **Developer as a real, separate role** — previously registerable but with no login path and no dashboard at all; now built the same way as Host/Guest.
- **Three real KYC bugs found and fixed:** no field existed to enter a non-NIN ID's actual number (the system silently substituted NIN regardless); admin's document review only checked a person's *primary* role, missing

real credentials on a secondary role; and stored, pre-signed document URLs are structurally fragile, now replaced with a fresh signed URL generated live at the moment of viewing.

- **Property reference numbers and admin search** (187, 188) — every property now has a permanent PROP-000001-style reference, backfilled for all existing properties, with a real admin search by reference/title/owner. Also fixed: an application awaiting an owner's decision was previously invisible to admin's own query.
- **Real PIN reset** (189) — phone + NIN verification; did not exist in any form before this.
- **Real FAQ page** — written from CHS's own T&C and actual features, reachable from the previously-placeholder "Help & FAQs" menu entry.
- **Add-a-role without logging out, plus a role switcher** — Profile page now shows every real role as a tappable pill and links straight to "Add another role," which skips the phone/PIN re-verification entirely for an already-authenticated session.
- **Desktop splash screen fix and four new per-role color themes** (Host, Guest, Developer, Staff) — a real, deliberate 7-zone color system already existed; these four roles were quietly borrowing another role's colors and now have their own.
- **Team/Staff subscription revenue model** (190, 191) — free for 2 staff, real subscription from the 3rd, tiered by staff count, with exact multi-month discount math (6 months paid → 8 months access; 12 paid → 18 access).

Supabase Status (confirmed at time of writing)

Every migration through 222_enable_realtime_notifications is live in the real, production Supabase project — checked directly against Supabase's own migration history, not assumed from local files. No drift between the exportable backend-v2/ folder and the live database. One real ERROR-level security advisory is outstanding (a SECURITY DEFINER view, public.public_profiles) and worth a follow-up look; the remaining advisories are standard hardening warnings, not active vulnerabilities.

New Since Last Handover Update (migrations 192–222)

- **Receipt/voucher redesign** — the real, live receipt page rebuilt with a genuine branded design (gradient header, watermark seal, serif amount display); no schema change, frontend only.
- **Artisan commission corrected** (201) — real, tiered 6%/8.5% by labor amount, replacing an earlier flat 10%. get_artisan_commission_pct() is the real source of truth; the old flat setting in platform_settings is dead, harmless data.
- **Complete marketplace rebuild** (196–197, 202–206) — admin-mediated, contact-filtered messaging (reusing detect_offplatform_contact); real escrow-backed payment at 6%/4% (accept_marketplace_quote, release_marketplace_escrow_to_vendor, refund_marketplace_escrow_to_buyer); a real direct-buy path (buy_product_direct) bypassing negotiation entirely. The old referral_fee_settings/referral_fees_owed flat-fee tables are now fully dead — nothing in current code reads from them.
- **Sister marketplace link** (207) — a real, admin-editable URL (get_sister_marketplace_link, reusing the contact-settings pattern), shown on Marketplace, Construction Roadmap, and Owner → Maintenance. **Still holds a placeholder URL — must be updated by admin to the real address.**
- **Rental application rework** (208) — real applicant_full_name, employer_business_name, employer_business_address columns; owner's review card rebuilt to show every real applicant field, not just the guarantor's.
- **Independent guarantor verification** (211) — real, complete redesign. submit_rental_application now takes only the guarantor's name/phone; a real, single-use token (GTN-...) is generated and the guarantor completes everything else about themselves at /guarantor-confirm/[token], requiring no CHS account.

A real `awaiting_guarantor_confirmation` status genuinely blocks the application from reaching the owner until this is done.

- **Event Centre real pricing** (212–214) — new `event_capacity_tiers` and `event_facilities` tables, host-manageable at `/event-pricing/[propertyId]`; `request_event_booking` computes and charges the real total (tier + facilities + 6%) in one call. A real, verified reference (the project's earlier prototype `index.html`) was checked directly before rebuilding this. Simple event-day add-on flags (`band/caterer/ushers/notes`) also added directly to `shortlet_bookings` for the standard booking flow.
- **Real payment regressions found and fixed** (215–218) — `pay_rent` and `pay_rent_to_own_installment` both silently omitted the payer's own real commission from the actual amount charged; both rewritten to compute and charge the true total in one atomic call, and both return a real JSON breakdown (`reference`, `real_total_paid`, etc.) the frontend now displays before payment.
- **Stale commission early warning** (219) — `get_stale_uncollected_commissions()`, surfaced as a real, visible alert on Admin → Finance for any commission invoiced but genuinely uncollected 2+ hours — the concrete safeguard against the above regression pattern recurring silently.
- **Clickable, real-time notifications** (220–222) — `notify_admins_by_domain` widened to accept a real `p_link` (it never did before); real links threaded through the full rental-application decision chain; **the notifications table is now genuinely in Supabase's real-time publication** (it never was before — this was the actual, confirmed cause of "no admin pop-up"). `NotificationBell.tsx` now subscribes via `postgres_changes` and shows a real, visible toast on arrival.
- **New real, dedicated pages:** `/my-applications` (buyer/tenant, all applications+offers), `/owner-applications` (owner, all received applications+offers, most recent first), Admin → Platform Earnings tab (real payer/amount/timestamp). Rent Savings widget removed from the general homepage — confirmed the tenant dashboard's own `WalletQuickView` already covered it; was a genuine duplicate, not a missing feature.

New Since Last Handover Update (migrations 223–274)

This was the single largest batch of the engagement — a long, iterative round of real client testing that surfaced genuine bugs across nearly every part of the app, plus several substantial new features built in direct response to real, repeated client requests. Grouped by theme rather than strict migration order, since many of these interleaved:

- **Notification click-through — three real attempts, the third one correct.** The underlying problem was never "no link exists" — it was how the click was wired. Attempt one (a `Link` wrapping a clickable `div`) was structurally fragile. Attempt two (`window.location.href`) fixed the click but caused a genuine regression: a full, disruptive app reload that lost in-progress admin work. The real, final fix is `router.push()` combined with explicit `stopPropagation/preventDefault` on the click handler — a proper Next.js soft navigation, not a hard reload. `NotificationBell.tsx` is the one shared component; fixing it once covers every dashboard.
- **Admin mediation extended to rental applications and offers, both directions** (235, 238, 243) — closing a real, serious gap: a buyer's offer, and separately an owner's accept/decision on that offer, both used to reach the other party directly with zero CHS review. Real `awaiting_admin_review / owner_decided_pending_relay` statuses now gate both directions, matching the same real pattern the marketplace already used. `admin_relay_application_to_owner`, `admin_relay_offer_to_owner`, and `admin_relay_offer_decision_to_buyer` are the three real relay functions — all now audit-logged (see below).
- **A buyer's real phone number, name, occupation, and source of funds** (233) — an offer previously required no real bio-data at all. Now required at submission, shown to admin for verification, and deliberately kept off the owner's own screen — contact stays mediated through the real, contact-filtered message thread, the same principle already enforced in the marketplace.
- **Admin document review, the single most serious gap found this round** (244–251) — the screen where admin approves a new listing showed only the title and location, with zero visibility into the real uploaded

legal documents, on a completely separate tab from where those documents could even be viewed. Fixed properly: every real document now shows directly on the approval card, with its own view link and verify/reject action, and — for Sale properties specifically — the "Verify" button is now genuinely disabled, not just a visual warning, until every document is verified and the owner's own ID is confirmed verified too.

- **Land and other non-residential categories showing a bedroom count** — the display only ever checked whether a database value was non-null, never whether the category should have one at all. Fixed with a real, shared `shouldShowBedrooms()` safeguard, and cleaned 36 real listings that had this bad data.
- **Real room/bedroom/toilet dropdowns, and a genuinely repeatable "Others" facility field** — replacing plain number inputs; a new, separate toilets column (previously conflated with bathrooms); a new `total_rooms` field; the already-existing but never-wired `other_facilities` jsonb column finally put to real use.
- **A genuinely serious "processed history" gap** — the real backend function for this existed from an earlier round but was never actually connected to any real admin screen; approved/rejected items simply vanished from view with no way to refer back. Extended to cover property listings and registrations (not just applications and offers), added real decision timestamps, and — critically — built the actual missing screen (`get_admin_processed_history()`, a new "Processed History" admin tab).
- **Audit Trail** (253–274, ongoing) — a real, permanent `audit_log` table with no update/delete permission for anyone, by design, plus a reusable `log_audit_event()` function. Wired into every real high-stakes action found so far: property verification/rejection, dispute rulings, escrow fund release, account suspension, all three admin relay functions, agent referral payouts, all three marketplace moderation actions, and agent management changes (link/revoke/replace). Genuinely not "every significant action" in the strictest sense yet — this is real, ongoing coverage, not a finished claim.
- **Feature Explainer** (ongoing) — a real, reusable `InfoTip` component, mobile-first (tap opens/closes it; hover also works on desktop), live on 16+ real, previously-confusing terms across buyer, owner, and tenant screens (commission, escrow, guarantor, CHS Agent ID, rent auto-pay, the difference between move-in/move-out condition reports, and others). Real, working foundation; extending coverage across the rest of the app's 100+ features continues as ongoing work.
- **Property manager listing access** — a real, confirmed gap: agents could list a property, managers genuinely could not. Fixed to match the identical pattern already used for agents; no separate database-level restriction ever existed, this was a frontend role-gate that simply never included manager.
- **Real room-video system, a cost-free alternative to a paid virtual-tour service** (264–274) — new `property_videos` and `video_requests` tables. An owner/agent/manager records or uploads a short, real video per room, on both new listings and (added after an honestly-flagged gap) existing ones via `/edit-listing/[id]`. A buyer/tenant can request a specific additional video if unsatisfied, and can now track that request's real status on `/my-applications` rather than only getting a one-time confirmation. Found and fixed three further real gaps purely by exercising the feature end to end: the owner's pending-request view had no direct link to actually act on it; the real notification itself (the first thing an owner would tap) linked to the public property page instead of the real edit page; both fixed.
- **Document Site** — a new real admin page (`/admin/document-site`) bundling every current reference document (User's Guide, Terms & Conditions, Feature Catalog in two formats, this Handover Notes file, the Fix Tracker, the Property Manager Guide, and the System Documentation the client separately provided) directly into the app's `public/` folder, so they're always available without depending on a separate storage service.
- **New Round 2 demo accounts** (phone range 0812 . . . , PIN 123456) — the original demo accounts were exhausted/mixed together across extensive testing. Five fresh accounts: an Owner (Land/House/Office/Warehouse listings), a Host (Chalet/Hotel/Event Centre, the Event Centre with real capacity tiers seeded), and Buyer/Tenant/Guest each funded at a real ₦100,000,000.

- **A genuine, real bug in how these Round 2 accounts were first created** — inserting directly into `auth.users` via raw SQL left several fields (`confirmation_token`, `recovery_token`, and related) as `NULL` where Supabase's real auth service expects an empty string, a documented pitfall of bypassing the official Admin API. Silently broke login for all five accounts. Found by a direct, full-field comparison against a known-working account; fixed by correcting the fields and re-hashing the password at the matching bcrypt cost factor.
- **A more saturated, "solid" color palette** — the client's core brand colors (charcoal + burnt amber) already matched a professionally-produced reference palette's own "best choice" option almost exactly; the real gap was that the individual per-role zone colors were noticeably more muted by comparison. All 14 zone colors upgraded to bolder tones drawn from the same reference.

Supabase Status (confirmed at time of writing, this update)

Every migration through 274 is live in the real, production Supabase project. No drift between the exportable backend-`v2/` folder and the live database. The same one real ERROR-level security advisory from the last update (`public.public_profiles`, a `SECURITY DEFINER` view) remains outstanding and still worth a follow-up look.

A note on process, worth reading before picking up this project

Several real mistakes happened this round, and each one was caught and corrected the same way: by actually testing against real data and the real, full build, not by trusting that a change compiled or looked right. A JSX edit that accidentally deleted a needed line; a React state-in-effect warning; a wrong TypeScript type for a Postgres join that returns an array, not a single object; a genuinely garbled shell heredoc that skipped packaging a zip entirely. None of these were caught by writing careful code the first time — they were caught by a disciplined habit of verifying afterward: full lint, full production build, direct database queries against real inserted test data, and — after one packaging mistake — actually re-opening the finished zip and searching it for the real code before calling anything done. Keep that habit. It's the actual reason "done" has held up more often than not.

New Since Last Handover Update (migrations 275–410)

Real-device testing dominated this round, not new-feature building — most fixes below were found by the client actually using the app, not by review.

- **Identity verification, rebuilt properly.** The original form collected an ID type and number only. Now collects the name exactly as printed on the ID (compared automatically against the registered name for the reviewing admin), gender, age bracket, state, address, occupation, email, and a required phone number, with explicit consent. Made mandatory before any commitment anywhere on the platform (offers, rental applications, shortlet/hire bookings, inspections, rent-to-own, marketplace purchases/quotes, Engage CHS), enforced at the database level via one shared trigger (`enforce_verified_identity`), not only on screen.
- **One shared rulebook for every phone, ID, and account number the platform collects** (`lib/validators.ts` on the frontend, matching functions in the database) — built after a direct client test deliberately submitted a 10-digit NIN and it was accepted. Checked live as you type, again on submit, and, non-negotiably, again in the database.
- **Rent escrow**, matching the protection buyers and shortlet guests already had. Rent now sits held in the landlord's wallet, released automatically on a genuinely clean move-in report, or after a 14-day grace period, with admin able to release early. The grace-period auto-release itself was a gap found late in the round (promised, never scheduled) — now a real daily cron job, tested against constructed clean and dirty tenancies.
- **A real, serious payment regression, found, reversed, and traced twice.** A duplicate rent payment was traced to a missing "too early" check; the fix then wrongly blocked brand-new tenancies' first payment, traced and fixed again. Separately, a tenant was able to pay a rental commission with the real rent never moving —

traced to an earlier round's own mistake (the wrong, generic commission function wired onto rentals) and fixed at the source.

- **The splash-screen complaint, open across several rounds, finally traced to its real, final cause.** Not the service worker (though that had its own, separate real regression and was removed outright) — the actual remaining cause was a notification-click fix that opened links in a literal new browser tab, which cannot avoid a full cold start. Reverted to same-tab navigation.
- **A serious data-protection gap, found while answering a direct client question, half-closed.** Owners could read an applicant's or buyer's complete raw record through the API — NIN, ID document, both phone numbers, guarantor details — despite the screen showing less. Protected views now exist and the screens use them; the old, unrestricted access is deliberately still open pending confirmation the new screens work (see Known Gaps).
- **A broader security audit,** done unprompted while working on documentation: 25 database functions with full privileges had no caller check at all, reachable by anyone including people not signed in — confirmed directly, an unauthenticated call returned 35 real registration records. Closed at the database level; see Known Gaps for the one exposed secret that still needs rotating.
- **A real, dedicated "Escrow Oversight" admin tab,** combining property-sale escrow, marketplace escrow, and shortlet/hire deposits into one place with a real combined total — none of this existed as a single view before.
- **Upload boxes redesigned everywhere** as real, visible drag-and-drop targets, after a direct complaint that a plain file input looked like a line of instruction text, not something clickable.
- **A "Take a Tour" photo/video gallery,** built for every property, ending in a genuine "satisfied, proceed" or "not for me" choice — with real photos (the client's own, once a hosting attempt through Claude's own tools genuinely failed and a direct-to-Supabase upload was used instead).
- **A complete rental-data reset,** on direct instruction, done precisely: every real wallet impact reversed first and calculated (not guessed) from the actual records, verified afterward against a known clean baseline account.

Supabase Status (confirmed at time of writing, this update)

287 real migrations are live in production (through version 20260928182240). backend-v2/ for migrations 383–410 was corrected this round to contain real, runnable SQL, verified directly against the live database; the 196–382 backlog was found to be comment-only and has not yet been corrected (see Known Gaps). Two ERROR-level security advisories remain, both correct-by-design SECURITY DEFINER views built specifically to bypass RLS safely inside a controlled function (owner_rental_applications, owner_offers).

- **Every real reference document in one tap:** Admin → 📁 Document Site (/admin/document-site) — User's Guide, Terms & Conditions, Feature Catalog (both formats), this Handover Notes file, the Fix Tracker, the Property Manager Guide, and System Documentation.
- **Every real feature, cross-referenced to its admin location:** CHS_COMPLETE_FEATURE_CATALOG.pdf, the segmented .xlsx, or in-app at /admin/feature-catalog (always current, same source data as the PDF).
- **How to use the app, by role:** CHS_USERS_GUIDE.docx or in-app /guide.
- **Resolving a specific user's support issue:** Admin → 🔍 Trace an Account — search by phone/email/name, see their full wallet, promotion, roadmap, and Engage CHS history in one place.
- **What a sub-admin did and when:** Admin → Overview → "View sub-admin action history."

New Since Last Handover Update (migrations 411–475f, October 2026)

Standing rules the client has set (apply to all new work): (1) a correction in one category goes across the board; (2) newest first everywhere — except upcoming due dates and scheduled inspections; (3) errors are fixed honestly, not with reassurances; (4) no test may move real data — tests run in transactions that are rolled back on purpose; (5) backend -

v2 is documentation and needs no permission; (6) CHS relays every request and message, counterparties never deal directly, and no counterparty phone number reaches the other side; (7) EVERY payment is made through the CHS Wallet and nobody pays an account claiming to be the platform; (8) prepaid principle: the host or seller is paid on arrival or confirmation; (9) every mortgage payment passes through CHS hold with an audit trail and the final payment is held until the property documents are handed over; (11) EVERY recorded event shows a full date AND time to the second in Nigerian time (WAT) using `formatDateTime()` in `lib/format.ts` — documents, confirmations, notifications, transactions, applications, admin actions; never a bare date for an event (calendar dates such as a lease end use `formatCalendarDate()`); (10) the final mortgage release is ALWAYS manual by a super admin unless a super admin has ticked the auto-release setting (off by default).

What was built (see each migration's header comment in backend-v2/):

- 411–439: automated verification system (AVS) for identity and guarantors; notification-link and join-bug fixes; commissions deducted at source; stale-offer tracker; accounting-grade notifications.
- 440–449: refund policy for every payer type; ledger hardening; Terms versioning with an acceptance log (`terms_current_version`, `accept_terms()`).
- 450–459: hotel / lodge / shortlet / event-centre booking: room units with a database-level double-booking guard, host calendar console, request-first booking, everything through CHS relay and message review, Terms Version 3, arrival confirmation with prompt host payout.
- 460–467: Rent to Own exact ownership and no overpayment; one active agreement per property; photo captions; inspection transport cost paid by the requester through the wallet; requests go through admin first; marketplace alerts, stock and orders; sale proceeds released exactly, seller can ask for release.
- 468–471: wallet security (transaction PIN, limits, 24-hour withdrawal pause on money received or added by card, device checks, SMS codes, freeze, fraud report, pattern scanner); Terms Version 4; the buyer's delivery phone is hidden from the seller.
- 472–474: hotel Instant Confirm, Express and peak pricing; host cancellation strikes (warning, 10% penalty, suspension within 90 days); digital check-in with arrival pass, registration card, lockbox code and front desk.
- 475 / 475c–475f: Mortgage through CHS. Owner decisions are relayed by admin; the applicant's name and phone are hidden from the owner (shown as "Alex P."); any amount can be paid; every instalment is held and released by admin; the final payment is held until a handover completes (buyer request, owner uploads documents, admin approves, owner sends with proof, buyer confirms). Final release is by a super admin only. `platform_settings.rto_final_auto_release` (default `false`) can be ticked by a super admin in the admin Rent-to-Own panel; `admin_revert_rto_final_release()` returns a released final payment to CHS hold.
- Front end: notification bell on every dashboard (`NotificationBell`, `lib/roleHome.ts`), newest-first ordering across dashboards, admin feed buttons that always navigate (`?_n= stamp`).
- Documents: Terms Version 5 (mortgage held by CHS end to end), Users Guide sections 16.8, 16.9, 17, 19.6, 19.7, 23; Feature Catalog section 17; FAQ and Privacy Policy updated.

Supabase status (October 9, 2026): 363 migrations recorded live (latest version 20261008161505), 602 public functions, 115 public tables. `platform_settings.terms_current_version` is 6 (Version 6 adds term 37, Records and timestamps).

Round 3 test accounts (October 9, 2026), phone login, PIN 123456: 08130000001 Event Centre Host, 08130000002 Event Centre Guest, 08130000003 Hotel & Lodge Host, 08130000004 Hotel & Lodge Guest (both guests hold ₦100,000,000). The event centres are Grand Hall (Test) and Royal Garden (Test), each with three capacity tiers and eleven priced facilities; the hotel is Savannah Grand Hotel (Test) with Standard, Executive and Family Suite room types; the lodge is Hilltop Lodge (Test) with Instant Confirm (the host must first tap "My calendar is accurate"). Guests can request ushers, a caterer, a live band, security and free-text extras on any event centre.

Switch-on checklist before real launch: set `transfer_requires_pin` and `withdrawal_requires_pin` to `true`; set `otp_step_up_enabled` to `true` only once the Termii SMS key works; rotate the SMS shared secret; keep

rto_final_auto_release off unless a super admin chooses otherwise for a busy period.

Known limits: earlier mortgage instalments paid under the old pay-the-owner-directly rule were not clawed back. There is no wanted-items bidding board (a vendor's "My Recent Bids" shows quote responses). The Vendor and Service Provider test data (476_round5_marketplace_test_data.sql) and the three test guides (Hospitality walkthrough, Vendor/Service Provider, Rent-to-Own) are written but have not yet been run end to end by a person. Mortgage due dates, a minimum deposit and an agreement document remain unbuilt. Mortgage screens were verified by build and automated screen tests, not yet on the live site by the client for every path.

7. Standing Discipline Going Forward

Every future batch of work should:

1. Apply real migrations directly to the live Supabase database, and test with real data before considering it done.
2. Update CHS_PROGRESS_LOG.md with what was actually done.
3. Update this handover note if a new architectural pattern, gap, or critical fact emerges.
4. Update the Users Guide and Feature Catalog (both formats) if user-facing or admin-facing functionality changed.
5. Rebuild and verify (lint + full production build) before packaging anything for GitHub/Netlify.
6. **When testing a real function that's expected to fail, run the setup/funding step as a separate call from the failing call.** Supabase's implicit transaction batching means a later statement's error can silently roll back an earlier, successful update in the same request — caught more than once by verifying real database state after a test rather than trusting a query appeared to succeed.
7. **Before assuming a client-reported visual bug is fully fixed, get their explicit confirmation from a real device.** A compiled, built, and packaged fix is not the same as a confirmed fix for anything involving rendering — this class of bug has needed a second real pass before, and standing here waiting on that same confirmation again.
8. **When researching a real-world legal or regulatory requirement (documents, permits, compliance) for a feature, cite genuine, current sources and say so plainly — never invent a plausible-sounding list.** This mattered directly for the Sale legal-document requirements.

Hotel operations, tranche 1 (October 9, 2026)

Host dashboard → hotel_lodge listing → "Hotel operations" (components/HotelOperations.tsx): Housekeeping, Repairs, Money. Database objects are listed in backend-v2/475 note 475k. Finance report, ledger, expenses and voids are owner-only; staff may run housekeeping, repairs and record income. Crons: chs-hotel-thank-yous, chs-hotel-owner-summaries. Messages are in-app only until a WhatsApp Business provider is connected. Tranche 2 (menu, POS, inventory, recipes, assets, branded page) not started.

Hospitality Suite (October 9, 2026, night)

Screens: host dashboard (My team, Plans), /host/work (duties, reports, hotel operations for the person's role), Hotel operations → Import & sync, Admin → Hotel Controls → Hospitality plans & pilot. Database objects are listed in backend-v2/475 note 475l. LAUNCH CHECKLIST: (1) deploy this code (Terms V7 is already live in the database, so deploy promptly); (2) in Admin set Pro and Business prices and commission cuts (blank price = not on sale); (3) grant the pilot to up to 8 operators; (4) only after the pilot, turn the master switch ON. Automatic calendar refresh, event-centre module and phase 4 are not built. Terms V7 and Privacy 7A are not lawyer-reviewed; the data-processing wording needs a Nigerian lawyer (NDPA 2023) before the pilot.

Event day control (Oct 10, 2026): components/EventOperations.tsx on the host dashboard and Work page for event_centre listings; objects in backend-v2/475 note 475m.

Hotel Phase 4 (Oct 10, 2026): HotelServices.tsx (staff) and GuestRoomMenu.tsx (guest); objects in note 475n. Money on orders is paid to the hotel directly and posts to host_income_entries; no CHS commission. Public hotel page built: app/hotel/[id]/page.tsx via get_public_hotel_page (note 475o).

No hidden prices (Oct 10, 2026): note 475q; lib/listingRules.ts, ListingRiskFlags.tsx, ListingGapsNotice.tsx, changes to list-property, edit-listing, VendorListingForm, owner and admin pages. Contact numbers are stored privately (property_contacts) and not shown publicly. Terms Version 8 (terms 39 and 40) covers this rule and is live in the database. Availability confirmation, listing reports and trust signals: note 475r, components ReportListingButton, AdminListingReports, ListingAvailability. Copied-photo detection: lib/imageHash.ts and note 475s; covers new listings only (older photos have no fingerprints).

Event quotations (Oct 10, 2026): EventQuoteHost.tsx (host, inside Event day control) and GuestEventQuote.tsx (My Bookings); note 475p. Extras are paid to the venue directly, outside CHS escrow; recording a payment posts to host_income_entries centre events.